

Decision Log — Skylark BI Agent

Key Assumptions

- **Board schema is unknown at build time.** The two source files weren't available while building, so I designed against a plausible schema (Work Orders: client, sector, status, start/end dates, order value, location; Deals: client, sector, stage, deal value, expected close date, owner) and built a column-alias mapping layer so the pipeline tolerates the real board's actual column names without code changes, as long as they're reasonably close. *list_available_fields* is the escape hatch if the real data differs meaningfully.
- **Sector/stage/status taxonomies are unknown**, so categories are fuzzy-clustered from whatever values actually appear on the board rather than matched to a hardcoded canonical list.
- **"This quarter" / "this year"** default to the current calendar period relative to the server date, stated explicitly in the reply so the user can correct it.
- **Currency is assumed INR-first** (Rs., lakh/crore suffixes parsed), with \$ and plain-number fallback.
- **Read-only integration**, per the spec — no mutation calls are ever issued to monday.com.

Trade-offs

Decision	Chosen	Why
monday.com access	Direct GraphQL API	MCP would mean managing a second server process inside a single hosted app for zero capability gain here. Direct API keeps it one deployable process.
How the LLM sees data	Discrete aggregation tools that compute real numbers in Python	Prevents arithmetic hallucination and keeps token usage low. A <code>get_raw_records</code> fallback tool covers ad-hoc questions the fixed aggregations miss.
Category normalization	Dynamic fuzzy clustering, no hardcoded taxonomy	Real taxonomy was unknown ahead of time; a hardcoded list risks silently misclassifying real values.
Frontend	Streamlit chat UI	Built-in chat primitives + free one-click hosting fit the 6-hour budget and "testable without local setup" requirement.
Data freshness	Fetch-and-cache per session + manual refresh	Avoids refetching on every message in a multi-question conversation; webhooks are the production answer but out of scope for a 6-hour prototype.
Date parsing	Detect ISO (YYYY-MM-DD) explicitly before dayfirst fallback	Caught in testing: <code>dateutil</code> with <code>dayfirst=True</code> silently misreads unambiguous ISO dates (e.g. 2025-12-01 read as <code>day=12/month=01</code>).

What I'd Do Differently With More Time

- **Validate against the real CSVs** — everything here was validated with synthetic messy data (unit tests + an end-to-end smoke test); the alias table and category assumptions should be checked against the actual imported boards on first real use.
- **Webhook-based cache invalidation** instead of a manual refresh button.
- **Confidence surfacing on fuzzy-matched categories** so a bad merge is easy to catch.
- **Session persistence** (a small DB) so conversation history survives a page refresh.
- **Chart output**, not just text, for leadership-update mode.
- **Automated tests against a real sandboxed monday.com board** to catch API-shape drift.

Interpreting "Prepare Data for Leadership Updates"

I read this as: the agent should be able to produce a structured executive brief on request, not generate slides or files. The system prompt instructs the agent that when asked to prepare something for leadership, it should respond with headline metrics, 3–5 bullet insights, and an explicit caveats section — text a founder can paste directly into a slide, email, or board update — rather than a raw data dump. I chose this over building a PPTX/PDF generator because the highest-value part of “leadership update prep” for a founder is almost always deciding what the 3–5 things worth saying are, which is a reasoning task the agent is already positioned to do well; turning that into a formatted deck is a comparatively mechanical last step better left to whatever tool the founder already uses to build decks.